Solving the two equations gives

$$r_{\mathrm{m}} = \frac{1+\sqrt{5}}{4} d \ . \tag{8}$$

IV.

Reference solution:

For convenience, all heats below denote magnitudes.

Combine an engine and heat pump: the engine converts boiler heat to work, which drives the pump to absorb outdoor heat and deliver it indoors.

The boiler is the engine's hot source; a cold source can be indoors or outdoors. For the room, let the engine absorb Q_0 from the boiler and reject Q_{10} indoors. Then

$$\frac{Q_0}{T_0} - \frac{Q_{10}}{T_1} \le 0 \ . \tag{9}$$

The pump's hot source is indoors, its cold source outdoors. It absorbs Q_2 outdoors and rejects Q_{12} indoors. Then

$$-\frac{Q_{12}}{T_1} + \frac{Q_2}{T_2} \le 0 \ . \tag{10}$$

The room receives a total heat from the combined cycle of

$$Q_1 = Q_{10} + Q_{12} \ . \tag{11}$$

Adding (9)、(10) gives

$$\frac{Q_0}{T_0} + \frac{Q_2}{T_2} - \frac{Q_1}{T_1} \le 0 \ . \tag{12}$$

The same result follows if the engine uses outdoors as its cold source.

Energy conservation gives

$$Q_1 = Q_0 + Q_2 \ . \tag{13}$$

To deliver Q_1 directly to the room, the boiler must release

$$Q_0' = Q_1 \ . \tag{14}$$

Combining (12)、(13) gives the boiler heat for combined heating:

$$Q_0 \ge \frac{T_0\left(T_1 - T_2\right)}{T_1\left(T_0 - T_2\right)} Q_1 \ . \tag{15}$$

The fractional cost reduction is

$$\frac{Q_0' - Q_0}{Q_0'} \le \frac{T_2\left(T_0 - T_1\right)}{T_1\left(T_0 - T_2\right)} \ . \tag{16}$$